

Course Syllabus - Course Run Year

CSE 578: Data Visualization

Contact Information

Instructor (IoR): Samira Ghayekhlou,
Ph.D

Preferred Name: Dr. Samira

Pronouns: She/her/hers

Live Events: Tuesdays, 11:00am - 12:00pm Arizona (AZ) Time

**Graduate Student
Assistant (GSA):** Ruthwik Reddy Ratna

Preferred Name: Ruthwik

Pronouns: He/him/his

**Live Support
Sessions:** Fridays, 1:00pm - 2:00 pm Arizona (AZ) Time

**Graduate Student
Assistant (GSA):** Kanishk Tanotra

Preferred Name: Kanishk

Pronouns: He/him/his

**Live Support
Sessions:** Thursdays, 11:00am - 12:00pm Arizona (AZ) Time

**Graduate Student
Assistant (GSA):** Mounika Siripurapu

Preferred Name: Mounika

Pronouns: She/her/hers

**Live Support
Sessions:** Wednesdays, 10:00am - 11:00am Arizona (AZ) Time

Content Questions: [Ed Discussion](#)

Project or Assignment Questions: [Ed Discussion](#)

Slack Channel: Direct Link: asu-2231-cse578-24709.slack.com

Note: You must join/access this workspace using your ASURITE credentials.

Content Issues: Course "Report an Issue" tool (clickable link on every content page)

Technical Support: [Coursera Learner Help Center](#)

Note: Please make sure you are logged in so that support personnel recognize you as an ASU learner.

General Support: mcsonline@asu.edu

Note: When sending an email about this class, please include the prefix "CSE 578" in the subject line of your message. Please use this email address for questions that are private in nature. If it is a question that would benefit your classmates, and is not private in nature, please post in Ed Discussion.

Course Description

Visual representations generated by statistical models help us to make sense of large, complex datasets through interactive exploration, thereby enabling big data to realize its potential for informing decisions. This course covers techniques and algorithms for creating effective visualizations based on principles from graphic design, visual art, perceptual psychology, and cognitive science to enhance the understanding of complex data.

Specific topics covered include:

- *Data transformations*
- *Exploratory querying*
- *Statistical graphics*
- *Time series analysis*
- *Exploratory spatial data analysis*

Technologies covered include:

- *Tableau*

Learning Outcomes

Learners completing this course will be able to:

- *Develop exploratory data analysis and visualization tools using Python and Jupyter notebooks*
- *Apply design principles for a variety of statistical graphics and visualizations including scatterplots, line charts, histograms, and choropleth maps*
- *Combine exploratory queries, graphics, and interaction to develop functional tools for exploratory data analysis and visualization*

Estimated Workload/ Time Commitment Per Week

Average of 18 - 20 hours per week

Required Prior Knowledge and Skills

This course will be very challenging, and learners are expected to learn the necessary technologies on their own time.

Proficient Mathematical Skills and Theoretical Understanding

- Basic statistics
- Discrete mathematics
- Computer organization
- Architecture
- Data structures
- Algorithms in computer science

Strong Application Skills

- Apply high-level programming languages (e.g., C++, Java) and scripting languages (e.g., Python) to complete tasks

Proficient Experience

- High-level programming languages (e.g., C++, Java) and scripting languages (e.g., Python)

Technology Requirements

Hardware

- Standard personal computer with major operating system
 - Personal computer with 8 GB RAM or higher and an x86-64 CPU. Must be able to install virtual machines on this computer. Computers with ARM processors (or any other architecture) will not work.
- Reliable, strong Internet connection
- Webcam
- Microphone

Software/Other

- To complete coursework (e.g., assignments and projects), these applications/languages are required:
 - Jupyter Notebooks
 - Python

Textbook and Readings

At the graduate level, inquiry, research, and critical reading are part of the learning experience; however, this course does not have a required textbook. Any required readings are provided within or are accessible through the course of the [ASU Library](#).

For interested learners, the instructor of record recommends the texts:

Munzner, T. (2014). *Visualization Analysis and Design*. CRC Press.

Zafarani, R., Abbasi, M.A., & Lui, H. (2014). *Social Media Mining, An Introduction*. Cambridge University Press.

Course Schedule and Important Dates

Course teams will not be working on ASU's days off* and those are listed in the Course Schedule. Please review the [ASU Days Off](#) for more details.

Week/Title	Begins at 12:01 AM Arizona (AZ) Time	Ends at 11:59 PM Arizona (AZ) Time
Welcome and Start Here	January 5	January 15
Week 1: Introduction to Data Visualization	January 9	January 15
<i>*Martin Luther King Jr. Day</i>	<i>January 16, 2022</i>	
Week 2: Introduction to Statistical Graphics	January 16	January 22
Week 3: Multivariate Analysis	January 23	January 29
Week 4: Temporal Analysis	January 30	February 5
Week 5: Geographical Data Analysis	February 6	February 12
Week 6: Hierarchical Data Analysis	February 13	February 19
Course Survey	This will be updated in your course.	This will be updated in your course.
Week 7: Additional Tools Used for Data Visualization	February 20	February 26
Final Exam	February 20	March 1
Request for Faculty Review: MCS Portfolio Project Report Inclusion Request Optional, degree-seeking learner degree requirement	February 20	March 15
Faculty Feedback for the Review: MCS Portfolio Project Report Inclusion Request	March 16	March 29

Optional, degree-seeking learner degree requirement		
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Grades are due March 3, 2023. Please see the [ASU Academic Calendar](#) for additional information.

Assignment Deadlines and Late Penalties

Unless otherwise noted, all graded work is due on **Sundays at 11:59 PM Arizona (AZ) time**. For learners with accommodations through [Student Accessibility and Inclusive Learning Services \(SAILS\)](#) and/or the [Pat Tillman Veterans Center \(PTVC\)](#), please work with your SAILS consultant and/or PTVC Advocacy Team, Connect, and your instructor.

Graded Quizzes

A daily-automatic late penalty of 5% is applied after the scheduled due date and time.

- **Week 1 Graded Quiz** - due at the end of Week 1
- **Week 2 Graded Quiz** - due at the end of Week 2
- **Week 3 Graded Quiz** - due at the end of Week 3
- **Week 4 Graded Quiz** - due at the end of Week 4
- **Week 5 Graded Quiz** - due at the end of Week 5
- **Week 6 Graded Quiz** - due at the end of Week 6

Assignments

A daily-automatic late penalty of 5% is applied after the scheduled due date and time.

- **Assignment 1: Dino Fun World** - due at the end of Week 1
- **Assignment 2: Graphing Dino Fun World** - due at the end of Week 2
- **Assignment 3: Dino Fun World Analysis** - due at the end of Week 3
- **Assignment 4: Dino Fun World Time Series Analysis** - due at the end of Week 4
- **Assignment 5: Geographic Data Analysis** - due at the end of Week 5

- **Assignment 6: Hierarchical Clustering** - due at the end of Week 6

Project

A daily-automatic late penalty of 25% is applied after the scheduled due date and time.

- **Course Project Progress Report** - due at the end of Week 4
- **Course Project Final Report** - due near the end of Week 7 on Saturday, February 25, 2023, at 11:59 PM Arizona (AZ) Time.

Exam

A single-automatic late penalty of 100% is applied after the scheduled due date and time.

- **Final Exam** - available from Monday, February 20, 2023 at 12:01 AM Arizona (AZ) Time until Wednesday, March 1, 2023 at 11:59 PM AZ Time

Course Content

Each course in the MCS program is uniquely designed by expert faculty, so learners can best master the learning outcomes. As a result, course features and experiences are not the same across all MCS courses. Learners are expected to plan accordingly to accommodate for these differences.

Feedback Descriptions

The feedback descriptions are specific to auto-graded or auto-feedback items in the course.

- **Limited:** you will be able to see your Total Score, which includes the overall total percent (%) and the number (#) of points.
- **Partial:** you will be able to see your Question Score, which includes the correct or incorrect status and the total points for each question.
- **Full:** you will be able to see your Options and Feedback, which includes any itemized additional feedback.

Content and Assessment Details

If you have specific questions related to instructional and assessment items in this course that you would like to be considered to be addressed in the weekly Live Event hosted by the instructor, please clearly indicate your request in your Ed Discussion thread.

Lecture Videos

The concepts you need to know are presented through a collection of video lectures. You may stream these videos for playback within the browser by clicking on their titles or download the videos. Where available, you may download the individual slides that go along with the videos. To further support learning, all of the videos include transcripts and most include PDF lecture slides. Assignment videos and project-related videos do not have PDF lecture slides because they are not lectures and have associated documents specific to them.

A media guide is included at the beginning of each week in the Overview section. These guides are designed to give you a snapshot description of each week's media components and to provide the week's PDF lecture slides or note-taking materials where available, so you can plan your learning and quickly go back and review material as you prepare for your coursework.

Readings

Suggested readings may accompany topics. They are supplementary or enrichment materials for you to further understand the course topics.

Discussions

Ed Discussion

Ed Discussion (Ed) is being used in place of Coursera Discussion Forums. The purpose of Ed Discussion is to provide a place for learners to ask questions and receive answers from course staff and peers about course content and course work. The course team is engaged in discussions, but it is also a space to clarify, support, and enrich learner-to-learner communication and learning. There are designated categories for course items. You must select a category and subcategory to start a thread.

Discussions in Ed are designed to provide:

- Clarification
- Feedback
- Enrichment and deeper learning

- Connections between concepts or key ideas
- Reflection opportunities of real-world experiences
- Respectful debate and perspective building
- Resource sharing
- Networking

Designated Assignment and Project Discussion in Ed Discussion

Use Ed to discuss items relating to the course assignments and project. Questions/Threads should be categorized by their designated title in Ed. Please check for questions already asked and answered, or marked as resolved.

In-Video Questions

Designed to support your learning, highlight specific content, encourage active viewing and/or note-taking, and provide practice opportunities, these are short, ungraded quizzes to test your knowledge of the concepts presented in the lecture videos. You may take your time, review your notes, and learn at your own pace because in-video questions are untimed. With unlimited attempts, you may retake these as often as you would like at any point in the course.

There are no late penalties. In-Video questions are not counted toward your final grade in the class.

Knowledge Checks

Designed to support your learning, these are short, ungraded quizzes to test your knowledge of the concepts presented in the lecture videos. You may take your time, review your notes, and learn at your own pace because knowledge checks are untimed. With unlimited attempts, you may retake these as often as you would like at any point in the course. You are encouraged to read the full feedback, review your answer choices, and compare them to the correct answers. With the feedback as your guide, you may use these as opportunities to study for other assessments and tasks in the course.

There are no late penalties. Knowledge Checks are not counted toward your final grade in the class.

Practice Quizzes

There is a practice quiz to help prepare you for each graded quiz. You may retake these as often as you like at any point in the course. You are encouraged to read the full feedback, review your answer

choices, and compare them to the correct answers. With the feedback as your guide, you may use these as opportunities to study for other assessments and tasks in the course.

There are no late penalties. Practice quizzes are not counted toward your final grade in the class.

Graded Quizzes

Weeks 1-6 each include one (1) graded quiz for a total of six (6) graded quizzes in the course. Each graded quiz includes ten (10) single-select or multiple-select multiple choice questions. You will be allowed one (1) attempt for each of these quizzes. There is no time limit on how long you take to complete each attempt at the quiz. Once you open a graded quiz or test, the timer will start and you are to complete the assessment in a single session. Graded quizzes in this course include partial feedback. Read the Graded Quiz and Exam Policy for your course for more information.

Graded quizzes count toward your final grade in the class.

Individual Programming Assignments

This course includes six (6) individual programming assignments. All assignments are provided in the first week of the course in the *Welcome and Start Here* section, so you can preview what is expected and design your own learning schedules to complete these on time. Each assignment is included in the graded coursework section at the end of the week in which it is due. You will complete these assignments in a Lab environment (Jupyter Notebooks).

Assignments count toward your final grade in the class.

Individual Project

This course includes one (1) individual project. All project deliverable descriptions are provided in the first week of the course in the *Welcome and Start Here* section, so you can preview what is expected and design your own learning schedules to complete these on time. Each project deliverable has a submission space at the end of the week it is due. As a set of one (1), the project may be included in the Request for Faculty Review: MCS Portfolio Project Report Inclusion Request, which is optional and for degree-seeking learners only.

The project counts toward your final grade in the class.

Practice Peer Reviews

There are three (3) optional peer reviews in this course, each associated with an activity so you can receive feedback on the deliverables you create for these practice assessments. Although optional, you are encouraged to complete these peer reviews so that everyone can benefit from feedback on their work. You are expected to review your peers' work in a professional and respectful manner. Although there are no late penalties, it is important to submit your work as close to the due date as you can because classmates grade most of the assignments within three (3) days of the due date, and if you submit yours too late, there may not be anyone to review your work.

Graded Peer Reviews

There are five (5) required peer reviews in this course, each associated with an individual programming assignment so you can receive feedback on the deliverables you create for these graded assessments. Although they are worth 0% of your grade, you are required to complete these peer reviews so that everyone can benefit from feedback on their work. You are expected to review your peers' work in a professional and respectful manner. Although there are no late penalties, it is important to submit your work as close to the due date as you can because classmates grade most of the assignments within three (3) days of the due date, and if you submit yours too late, there may not be anyone to review your work.

Request for Faculty Review: MCS Portfolio Project Report Inclusion Request

This is an optional task for degree students wanting to use this course's project(s) as part of their portfolio degree requirement/specialization requirements. Review your onboarding course and the Welcome and Start Here section of your course for more details. The submission space is towards the end of the course.

Although there are no late penalties, these requests must be submitted by the designated deadline. The Request for Faculty Review: MCS Portfolio Project Report Inclusion Request does not count toward your final grade in the class.

- Address these project components in your Request for Faculty Review: MCS Portfolio Project Report Inclusion Request:
 - Course Project Progress Report
 - Course Project Final Report

Practice Exams

In order to help you prepare for your proctored exam, you have three (3) practice exams. Since they are intended to be practice opportunities and to help you learn, they are untimed, ungraded, and include feedback. You may engage with your peers in Ed Discussion to address questions, share resources and strategies, and provide feedback to help one another learn. You are encouraged to read the full feedback, review your answer choices, and compare them to the correct answers. You are encouraged to submit questions in Ed Discussion for the course team to address during Live Sessions. Use the feedback to guide your learning and to study for the proctored exam.

There are no late penalties. Practice exams are not counted toward your final grade in the class.

Proctored Exam

You have one (1) proctored, timed exam. Proctored exams include limited feedback. Read the Graded Quiz and Exam Policy for your course for more information.

No late exams will be permitted or accepted and will result in a score of zero (0) points. This does not include established accommodations for learners receiving accommodations through [Student Accessibility and Inclusive Learning Services \(SAILS\)](#) and and/or the [Pat Tillman Veterans Center \(PTVC\)](#). Proctored exams count toward your final grade in the class.

Exam Details	Final Exam
Content Covered	Weeks 1, 2, 3, 4, 5, and 6
Question Type	Single-answer and multiple-answer multiple choice questions
Number of Questions	31 total questions (30 content questions + 1 academic integrity question)
Availability Start	Monday, February 20, 2023 at 12:01 AM AZ Time
Availability End	Wednesday, March 1, 2023 at 11:59 PM AZ Time

Last Available ProctorU Appointment	Wednesday, March 1, 2023 at 9:01 PM AZ Time
Duration	120 minutes + plan for at least 15 minutes for proctoring set up

Exam Allowances

- Any resources not included in this list are **not** allowed during the exam or in your exam space.
- **Calculators:**
 - Four Function Calculator
 - Scientific Calculator
 - Computer's Calculator (basic or scientific mode only)
- **Notes:** Please read the specifications and plan ahead accordingly.
 - Handwritten notes: Hard copy; two (2) sheets; 8.5"x 11" or international A4 paper; double-sided.
 - Digital and typed/word processed hard copy notes will **not** be allowed unless there is an accommodation on file with SAILS.
- **Web:** Course website (to access exam only)
- **Whiteboard, scratch paper, writing utensils, erasing resources:** Learners are *strongly* encouraged to use the whiteboard option instead of scratch paper.
 - If using a whiteboard, learners may have erasable whiteboard markers and what is needed to erase writing on the whiteboard; please have extra whiteboard markers and eraser resources in your testing area.
 - If using scratch paper, learners may have an unlimited amount of blank scratch paper of any size, writing utensils (e.g., pens, pencils, markers, and/or highlighters) and erasers; please have extra ones in your testing area should you run out of ink, the pencil breaks, etc.
 - Before the exam concludes and the proctoring session ends, all scratch paper must be destroyed and all whiteboard markings must be erased. The last

question in the exam will be a confirmation of learners executing these ASU academic integrity actions.

- **Other:**
 - Learners are to independently take the exam in a single session without leaving the testing space (e.g., no bathroom breaks) to ensure proctoring of the entire session. Once you open the exam, your testing session begins.
 - You will be allowed one (1) attempt to take and complete each exam.
 - Learners are to stay within a clear view of the proctor throughout the duration of the proctored exam session.
 - You will be unable to open the exam until the exam proctor enters the password during the date and time you scheduled to take your exam with [ProctorU](#).
 - Your exam will automatically be submitted if it is not completed before the deadline.
- **Reminder:** All virtual machines must be closed *prior* to starting proctoring.

Proctoring

[ProctorU](#) is an online proctoring service that allows learners to take exams online while ensuring the integrity of the exam for the institution.

- You are expected to scan your testing space using your webcam for the proctor. Proctoring also requires you to have adequate sound and a working microphone. Please plan accordingly.
- You are strongly encouraged to schedule your exam(s) within the first two weeks of the course to ensure you find a day and time that works best for your schedule. Time slots can fill up quickly, especially during high volume time periods.
 - You *must* set up your proctoring at least 72 hours prior to the exam.
- **The exam proctor will input the exam password.**
- Additional information and instructions are provided in the *Welcome and Start Here* section of the course.

- When you are going to schedule exams, you *must* pick “Coursera” as your institution.
- Learners with exam accommodations through [Student Accessibility and Inclusive Learning Services \(SAILS\)](#) and [Pat Tillman Veterans Center \(PTVC\)](#), should not schedule exams until they receive an email invitation specifically for them from ProctorU.
- Your ID needs to be in English. See your MCS Onboarding Course for more information.

Course Grade Breakdown

Course Work	Quantity	Team or Individual	Percentage of Grade
Graded Quizzes	6	Individual	10%
Programming Assignments	6	Individual	30%
Assignment Peer Reviews	5	Individual	0%
Required Checkpoint: Project Preparation	1	Individual	0%
Course Project Progress Report*	1	Individual	15%
Course Project Final Report*	1	Individual	25%
Final Exam	1	Individual	20%

*The project(s) counts for 30% or more of the overall course grade, so this is a portfolio eligible course. See the [MCS Graduate Handbook](#) for more information about the portfolio requirement if you are a degree student.

Grade Scale

You must earn a cumulative grade of 70% to earn a “C” in this course. You must earn at least a “C” to receive graduate credit. This course has no grade curving. All graded coursework will be included to calculate grades (i.e., no graded items will be dropped). Grades will not be rounded. Grades in this course will not include pluses or minuses.

The instructor reserves the right to adjust individual grades based on, but not limited to: violations of academic integrity.

Letter Grade	Percentage
A	90% - 100%
B	80% - 89.99%
C	70% - 79.99%
D	60% - 69.99%
E	<60%

Live Events

This course has two types of live events: **Instructor Live Events** and **GSA Live Support Sessions**. Check the Live Events page in your course for your local time and access details. Although we try to be consistent for our learners' planning purposes, the Live Event schedule is subject to change throughout the course, so stay up-to-date on the event details by checking your Course Announcements and the Live Events page in your course.

You may join all live events from the course's Live Events page. The event's title will become active as a Zoom link ten (10) minutes before each event starts. You will also receive an email with a link to the Live Event or Live Support Session the day before the event starts.

Read about the specific policies related to Live Events in the Policy section of this syllabus: Live Events, Policy Regarding Expected Classroom Behavior, and the Student Code of Conduct for more detailed information.

Instructor Live Events - Weekly

Instructor Live Events are a valuable part of the learning experience because learners can meet with the course instructor and fellow classmates to learn more about course topics, special topics within the field, and discuss coursework. If you are able to attend these events, you are strongly encouraged to do so. If you have specific questions or topics of interest to be discussed during these events, please indicate your request in an Ed Discussion thread. Although it may not be possible to address all requests during the live event, the instructor is interested in tailoring this time to your questions and interests. The instructor will be following a set agenda, so please be mindful of that when engaging in the live event.

Instructor Live Events will be recorded and uploaded to the course. These can be found at the end of each week in the course.

GSA Live Support Sessions - Weekly

Live Support Sessions offer a chance for learners to get their questions answered from the course team. Although the course team is responsive to trends in Ed Discussion and mcsonline@asu.edu emails, these events focus on addressing learners' specific questions related to content: clarifications, reteaching, assessment review, etc. These sessions are not intended to address program or course design questions or feedback. GSAs do not have the authority to weigh in or make decisions regarding those items, so please do not include those at this time. These sessions are specific to helping learners learn materials and understand various course assessments. Feedback of that nature is best addressed in the communication channel: mcsonline@asu.edu and please include it in your course survey.

Live Learner Support Sessions are recorded, but not uploaded into the course.

Slack Channel

This course will have a unique Slack workspace where you can communicate with your classmates.

You must join/access this workspace using your ASURITE credentials.

While your instructors have access to the Slack workspace, it is intended to provide a space to create community with your classmates. Please remember to follow the communication protocol pinned in

your Slack channel to ensure that any questions or concerns you have are addressed in a timely manner. Also, please remember [ASU's Academic Integrity policy](#), and refrain from sharing assessment questions, answers or solutions.

Course Outline with Assignments

Please review the [ASU Days Off](#) for more details. Course teams will not be working on ASU's days off.

Welcome and Start Here (1/5 - 1/15)

Topics

- ☐ Academic Integrity
- ☐ Course Syllabus
- ☐ Required Prior Knowledge and Skills and Technology Requirements
- ☐ Course [Assignment(s) and/or Project(s)] Overview
- ☐ Peer Review Information
- ☐ Exam Information and ProctorU

Other Tasks

- ☐ Required Checkpoint: Technology Access and Installation
- ☐ Activity: Zeemap
- ☐ Discussion: Get to Know Your Classmates (in Ed Discussion)
- ☐ Schedule your proctoring with [ProctorU](#) for your proctored exam(s)
 - For learners needing accommodations, submit requests through [Connect](#) and review the [Student Accessibility and Inclusive Learning Services \(SAILS\)](#) and/or the [Pat Tillman Veterans Center \(PTVC\)](#).
 - Learners with exam accommodations through SAILS or PTVC should **not** schedule exams until they receive an invitation specifically for them from ProctorU.
- ☐ Required Checkpoint: Getting Started Quiz

Graded Coursework

- ☐ N/A

Week 1: Introduction to Data Visualization (1/9 - 1/15)

Topics

- ☐ Introduction to Data Visualization
- ☐ Introduction to Data Exploration Components
- ☐ Exploratory Querying and Visual Variables Used in Data Exploration and Visualization

Other Tasks

- ☐ Schedule your proctoring with [ProctorU](#) for your proctored exam(s).
 - For learners needing accommodations, submit requests through [Connect](#) and review the [Student Accessibility and Inclusive Learning Services \(SAILS\)](#) and/or the [Pat Tillman Veterans Center \(PTVC\)](#).
 - Learners with exam accommodations through SAILS or PTVC should **not** schedule exams until they receive an invitation specifically for them from ProctorU.
- ☐ Week 1 Practice Quiz

Graded Coursework

- ☐ Week 1 Graded Quiz
- ☐ Assignment 1: Dino Fun World

Week 2: Introduction to Statistical Graphics (1/16 - 1/22)

Topics

- ☐ Design Principles for the Most Widely Used Data Visualization Charts
- ☐ Design Principles for Box Charts and Q-Q Plots

Other Tasks

- ☐ Schedule your proctoring with [ProctorU](#) for your proctored exam(s).
 - For learners needing accommodations, submit requests through [Connect](#) and review the [Student Accessibility and Inclusive Learning Services \(SAILS\)](#) and/or the [Pat Tillman Veterans Center \(PTVC\)](#).

- Learners with exam accommodations through SAILS or PTVU, should **not** schedule exams until they receive an invitation specifically for them from ProctorU.
- ☐ Week 2 Practice Quiz
- ☐ Assignment 2 Peer Review: Graphing Dino Fun World Visualizations
- ☐ Required Checkpoint: Project Preparation

Graded Coursework

- ☐ Week 2 Graded Quiz
- ☐ Assignment 2: Graphing Dino Fun World

Week 3: Multivariate Analysis (1/23 - 1/29)

Topics

- ☐ Introduction to Multivariate Analysis
- ☐ Mosaic Plots and Pixel Based Displays
- ☐ Parallel Coordinate Plots
- ☐ Text Visualization
- ☐ Supervised Learning
- ☐ Unsupervised Learning

Other Tasks

- ☐ Week 3 Practice Quiz
- ☐ Assignment 3 Peer Review: Graphing Dino Fun Analysis Visualizations

Graded Coursework

- ☐ Week 3 Graded Quiz
- ☐ Assignment 3: Dino Fun World Analysis

Required Checkpoint: Weeks 1-3 Practice Exam (1/30 - 2/12)

Other Tasks

- ☐ Complete the practice exam

Week 4: Temporal Analysis (1/30 - 2/5)

Topics

- ☐ Introduction to Temporal Analysis and Visualization
- ☐ Strings and Sequences
- ☐ Edit Distance
- ☐ Introduction to W-Grams and Other Approaches
- ☐ Time Series
- ☐ Time Series Matching
- ☐ Time Series Motifs

Other Tasks

- ☐ Week 4 Practice Quiz
- ☐ Assignment 4 Peer Review: Time Series Analysis Visualizations

Graded Coursework

- ☐ Week 4 Graded Quiz
- ☐ Assignment 4: Time Series Analysis
- ☐ Course Project Progress Report

Week 5: Geographical Data Analysis (2/6 - 2/12)

Topics

- ☐ Introduction to Geographical Data Analysis
- ☐ Choropleth Maps
- ☐ Spatial Statistics
- ☐ Spatial Autocorrelation
- ☐ Spatial Scan Statistics
- ☐ Geovisual Analytics Systems

Other Tasks

- ☐ Week 5 Practice Quiz

- ☐ Assignment 5 Peer Review: Geographic Data Analysis Visualization

Graded Coursework

- ☐ Week 5 Graded Quiz
- ☐ Assignment 5: Geographic Data Analysis

Week 6: Hierarchical Data Analysis (2/13 - 2/19)

Topics

- ☐ Trees and Hierarchies
- ☐ What is Hierarchical Clustering?

Other Tasks

- ☐ Week 6 Practice Quiz
- ☐ Assignment 6 Peer Review: Hierarchical Clustering Visualization
- ☐ Complete the course survey before your final exam (strongly encouraged, appreciated, and used by the course team).

Graded Coursework

- ☐ Week 6 Graded Quiz
- ☐ Assignment 6: Hierarchical Clustering

Week 7: Additional Tools Used for Data Visualization (2/20 - 2/26)

Topics

- ☐ Tableau Basics
- ☐ Loading Data into Tableau
- ☐ Advanced Tableau
- ☐ Time Series Data Visualizations in Tableau
- ☐ Geographic Visualizations in Tableau
- ☐ Hierarchical Visualizations in Tableau

Other Tasks

- ☐ Tableau Peer Reviews:
 - ☐ Tableau Visualization Assignment: Line Graph
 - ☐ Tableau Visualization Assignment: Treemap
 - ☐ Tableau Visualization Assignment: Time Series
- ☐ Schedule your proctoring with [ProctorU](#) for your proctored exam(s), if you have not already done, at least 72 hours prior to your desired exam date and within the availability window.
- ☐ Request for Request for Faculty Review: MCS Portfolio Project Report Inclusion Request.
 - Optional for degree students wanting to use this course's projects as part of their portfolio degree requirement/specialization requirements.
- ☐ Complete the course survey before your final exam (strongly encouraged, appreciated, and used by the course team).

Graded Coursework

- ☐ Course Project Final Report (due Saturday, February 25, 2023, at 11:59 PM Arizona Time)

Final Exam (2/20 - 3/1)

Reminders

- ☐ Complete the course survey before your final exam (strongly encouraged, appreciated, and used by the course team).
- ☐ Schedule your proctoring with [ProctorU](#) for your proctored exam(s), if you have not already done, at least 72 hours prior to your desired exam date and within the availability window.
- ☐ Covers content from Weeks 1, 2, 3, 4, 5, and 6.
- ☐ Review the details and allowances information for this exam.
- ☐ Prepare for the exam and complete the practice exam.

Policies

All ASU and Coursera policies will be enforced during this course. For policy details, please consult the MCS Graduate Handbook and the MCS Onboarding Course.

Graded Quizzes and Exams

Each course in the MCS program is uniquely designed by expert faculty so that learners can best master the learning outcomes specific to each course. By design, course features and experiences are different across all MCS courses.

In the MCS program, we strive to provide learners with exercises and applied practice beyond quizzes and exams that align with the hands-on nature of the computer science industry. Ungraded practice opportunities *may* include, but are not limited to: in-video-questions (IVQs), knowledge check quizzes (KCs), weekly practice quizzes, practice exams, and other assignments or exercises. For all these learning activities, the questions and correct answers are provided to learners. When available, auto-generated typed feedback is built into the course to further help learners learn in real-time. Please thoroughly review your course to ensure that you are aware of the types of practice opportunities available to you.

For academic integrity purposes, once grades are made available, learners will see their overall total scores. Like other standardized tests, such as the GRE and SAT, learners will receive a singular grade for the graded quizzes and exams, but the questions, correct and incorrect answers, and feedback to each question will **not** be provided.

If learners desire 1:1 feedback for their questions on graded assessments, please submit questions to mcsonline@asu.edu. Rather than receiving the exact questions learners had correct and incorrect and the answers to those questions, learners will likely receive the concepts that were covered in the assessment questions so they will know what they need to review prior to other assessments and how to apply this information in their professional environments.

Absence Policies

There are no required or mandatory attendance events in this online course. Live Events, both Live Sessions hosted by the instructor and GSA Live Support Sessions hosted by the course team do not take attendance.

Learners are to complete all graded coursework (e.g., projects and exams). If exceptions for graded coursework deadlines need to be made for excused absences, please reach out to the course team by the end of the second week of the course using the mcsonline@asu.edu email address. Review the exam availability windows and schedule accordingly. The exam availability windows allow for your

own flexibility and you are expected to plan ahead. Personal travel does not qualify as an excused absence and does not guarantee an exception.

Review the resources for what qualifies as an excused absence and review the late penalties in the Assignment Deadlines and Late Penalties section of the syllabus and the course:

- a. Excused absences related to religious observances/practices that are in accord with [ACD 304–04](#), “Accommodation for Religious Practices” (please see [Religious Holidays and Observances](#)).
- b. Excused absences related to university sanctioned events/activities that are in accord with [ACD 304–02](#), “Missed Classes Due to University-Sanctioned Activities”.
- c. Excused absences related to missed class due to military line-of-duty activities that are in accord with [ACD 304–11](#), “Missed Class Due to Military Line-of-Duty Activities,” and [SSM 201–18](#), “Accommodating Active Duty Military”.

Live Event Expectations

The environment should remain professional at all times. Inappropriate content/visuals, language, tone, feedback, etc. will not be tolerated, reported and subject to disciplinary action. Review the Policy Regarding Expected Classroom Behavior section of the syllabus and the Student Code of Conduct for more detailed information.

Policy Regarding Expected Classroom Behavior

The aim of education is the intellectual, personal, social, and ethical development of the individual. The educational process is ideally conducted in an environment that encourages reasoned discourse, intellectual honesty, openness to constructive change, and respect for the rights of all individuals. Self-discipline and a respect for the rights of others in the university community are necessary for the fulfillment of such goals. An instructor may withdraw a student from a course with a mark of “W” or “E” or employ other interventions when the student’s behavior disrupts the educational process. For more information, review [SSM 201–10](#).

If you identify something as unacceptable classroom behavior in any communication channel (e.g., Ed Discussion, Zoom, Live Events, Slack, etc.), please notify the course team using the mcsonline@asu.edu email. For more specifics on appropriate participation, please review our Netiquette infographic.

Our classroom community rules are to:

- Be professional
- Be positive
- Be polite
- Be proactive

Academic Integrity

Students in this class must adhere to ASU's academic integrity policy, which can be found at <https://provost.asu.edu/academic-integrity/policy>). Students are responsible for reviewing this policy and understanding each of the areas in which academic dishonesty can occur. In addition, all engineering students are expected to adhere to both the ASU Academic Integrity [Honor Code](#) and the Fulton Schools of Engineering [Honor Code](#). All academic integrity violations will be reported to the Fulton Schools of Engineering Academic Integrity Office (AIO). The AIO maintains a record of all violations and has access to academic integrity violations committed in all other ASU colleges/schools.

Specific academic integrity announcements for this class:

- Computer science is a field that honors integrity to the maximum extent. This course has a zero-tolerance policy towards academic integrity violations.
- The recommended sanctions for violations are the failure of this course (no refunds) with a failing grade and reporting the violation to the university.

Copyright

The contents of this course, including lectures (Zoom recorded lectures included) and other instructional materials, are copyrighted materials. Students may not share outside the class, including uploading, selling or distributing course content or notes taken during the conduct of the course. Any recording of class sessions is authorized only for the use of students enrolled in this course during their enrollment in this course. Recordings and excerpts of recordings may not be distributed to others. (see [ACD 304–06](#), “Commercial Note Taking Services” and ABOR Policy [5-308 F.14](#) for more information).

You must refrain from uploading to any course shell, discussion board, or website used by the course instructor or other course forum, material that is not the student's/learner's original work, unless the

student/learner first complies with all applicable copyright laws; faculty members reserve the right to delete materials on the grounds of suspected copyright infringement.

Policy Against Threatening Behavior, per the Student Services Manual, ([SSM 104-02](#))

Students, faculty, staff, and other individuals do not have an unqualified right of access to university grounds, property, or services (see [SSM 104-02](#)). Interfering with the peaceful conduct of university-related business or activities or remaining on campus grounds after a request to leave may be considered a crime. All incidents and allegations of violent or threatening conduct by an ASU student (whether on- or off-campus) must be reported to the ASU Police Department (ASU PD) and the Office of the Dean of Students.

Disability Accommodations

Suitable accommodations will be made for students having disabilities. Students needing accommodations must register with [ASU Student Accessibility and Inclusive Learning Services](#). Students should communicate the need for an accommodation at the beginning of each course so there is sufficient time for it to be properly arranged. These requests should be submitted through the [online portal](#). See [ACD 304-08](#) Classroom and Testing Accommodations for Students with Disabilities. ASU Student Accessibility and Inclusive Learning Services will send the instructor of record a notification of approved accommodations and students are copied on these letters. It is recommended that students reply to the faculty notification letters, introduce themselves to their instructor, and share anything they might want to disclose.

Harassment and Sexual Discrimination

Arizona State University is committed to providing an environment free of discrimination, harassment, or retaliation for the entire university community, including all students, faculty members, staff employees, and guests. ASU expressly prohibits discrimination, harassment, and retaliation by employees, students, contractors, or agents of the university based on any protected status: race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, and genetic information.

Title IX is a federal law that provides that no person be excluded on the basis of sex from participation in, be denied benefits of, or be subjected to discrimination under any education program or activity. Both Title IX and university policy make clear that sexual violence and harassment based on sex is

prohibited. An individual who believes they have been subjected to sexual violence or harassed on the basis of sex can seek support, including counseling and academic support, from the university. If you or someone you know has been harassed on the basis of sex or sexually assaulted, you can find information and resources at <https://sexualviolenceprevention.asu.edu/faqs>.

Mandated sexual harassment reporter: As a mandated reporter, I am obligated to report any information I become aware of regarding alleged acts of sexual discrimination, including sexual violence and dating violence. ASU Counseling Services, <https://eoss.asu.edu/counseling>, is available if you wish to discuss any concerns confidentially and privately.

Disclaimer

The information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Course Creator(s)



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Ross Maciejewski (Dr. Ross) is an Associate Professor at Arizona State University in the School of Computing, Informatics & Decision Systems Engineering and Director of the Center for Accelerating Operational Efficiency, a Department of Homeland Security Center of Excellence. His primary research interests are in the areas of geographical visualization and visual analytics focusing on public health, dietary analysis, social media, criminal incident reports, and the food-energy-water nexus.



Huan Liu

Professor Huan Liu joined ASU in 2000 after conducting research in Telecom (Telstra) Australia Research labs and teaching at the National University of Singapore. He has extensive experience in research and development. Liu's research and teaching focuses on machine learning, data mining, and real world applications.



Selcuk Candan

K. Selcuk Candan is a professor of computer science and engineering at Arizona State University and the director of ASU's Center for Assured and Scalable Data Engineering (CASCADE). His primary research interest is in the area of management and analysis of non-traditional, heterogeneous, and imprecise (such as multimedia, web, and scientific) data.